



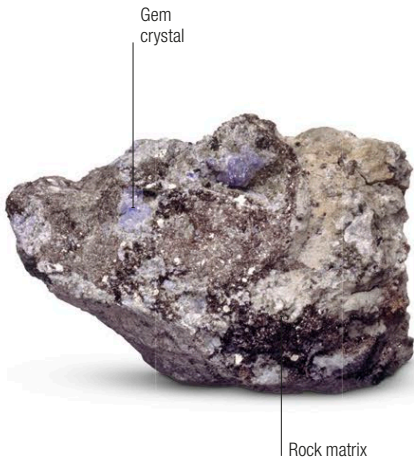
Haüyne

△ **Faceted haüyne** with a modified brilliant cut

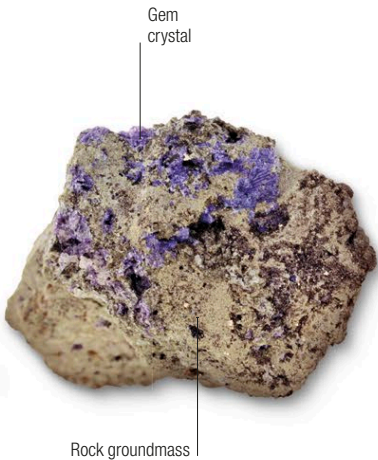
Haüyne is one of the components of lapis lazuli, along with pyrite, lazurite, calcite, and sodalite. Blue is the most common colour and occurs in lapis lazuli, but haüyne also comes in white, grey, yellow, green, or pink. Single crystals are sometimes found, and can be faceted only with difficulty – the mineral has perfect cleavage (planes of breakage), which makes it hard to cut without shattering the material. Facet-grade haüyne crystals tend to be small, with faceted stones usually weighing five carats or less.

Specification

Chemical name Sodium, calcium aluminosilicate with sulphate
Formula $\text{Na}_3\text{Ca}(\text{Al}_3\text{Si}_3\text{O}_{12})(\text{SO}_4)$ | **Colours** Blue, white, grey, yellow, green, or pink | **Structure** Cubic | **Hardness** 5.5–6 | **SG** 2.4–2.5
RI 1.49–1.51 | **Lustre** Vitreous to greasy | **Streak** Blue to white
Locations Germany, Italy, USA, Serbia, Russia, Morocco, China



Crystals in matrix | Rough | A number of small but good-quality, transparent patches of haüyne crystals are contained within this large rock.



Gemmy crystals | Rough | In this specimen a group of small, intensely coloured gemmy crystals of haüyne have developed in a rock groundmass.



Flawed gem | Cut | Gem-quality haüyne is so rare in pieces over one carat that even stones cut from slightly flawed rough are acceptable, as here.

René Haüy's name is one of only 72 that are inscribed on the Eiffel Tower

Superb colour | Colour variety | Although far from flawless, the stunning colour of this 0.82-carat, pear-shaped German gem makes it a highly desirable stone.



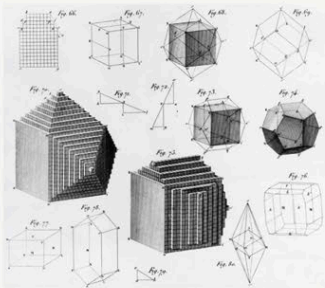
Internal flaws

Crown facet

Napoleon's professor

The father of crystallography

Haüyne is named after René Just Haüy (1743–1822), who grew interested in crystallography when he noticed that fragments of a broken calcite crystal cleaved along straight lines that met at constant angles. He was the first to show that a crystal is built up of tiny, identical units. In 1802 (after imprisonment in the French Revolution), Haüy was appointed Professor of Mineralogy at the Museum of Natural History in Paris. At Napoleon's request, he wrote a book on crystallography.



Illustrated crystals This plate is from *Treatise of Crystallography*, by René Just Haüy, 1822.